The Effects of Monetary Policy in Colombia Revisited: Policy Shocks vs. Information Shocks

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Abstract

This paper estimates the effects of monetary policy on output and inflation in Colombia, distinguishing between pure monetary policy shocks and central bank information shocks embedded in high-frequency policy surprises and financial market data, following Jarociński and Karadi (2020). To do so, the paper exploits high-frequency comovements between short-term break-even inflation expectations (BEI) and overnight index swap (OIS) rates around Banco de la República policy announcements. Pure monetary policy shocks are identified by a negative comovement between interest rates and inflation expectations, whereas central bank information shocks are characterized by a positive comovement. The paper then employs an Instrumental Variable Bayesian Vector Autoregression (IV-BVAR) framework, using the decomposed high-frequency surprises as external instruments to identify dynamic impulse responses. The results show that while raw monetary policy surprises generates a transient "price puzzle" in headline inflation, this anomaly fully disappears when instrumenting with the pure monetary policy shock, which triggers immediate, monotonic declines in both output and prices. Consistent with this, the information shock accounts for the short-run inflationary pressures of policy surprises.

Keywords: Monetary Policy Surprises, Monetary Policy Shocks, Information Shocks, Break-Even Inflation.
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What are the effects of monetary policy on the macroeconomy? This paper revisits this question for the case of Colombia distinguishing between pure monetary policy shocks and information shocks embedded in high-frequency policy surprises and financial market prices following Jarociński and Karadi (2020).

As has been long established[1], finding the answer to the longstanding question of the effects of monetary policy relies critically on the ability of researchers to isolate monetary policy shocks, that is, variations in the policy instrument that are plausibly not a response to the past, current, or future state of the economy itself. In principle, these shocks can be used to calculate the impact of pure policy variations on the economy. Over the years, different methods have been proposed to isolate these shocks. Among them, one that has become popular in recent times is to exploit the richness of high-frequency financial markets information (High-Frequency Identification, HFI) close and around those snapshots of time when Central Banks make announcements on policy decisions.

Assuming that the state of the economy is observed to the same extent by both the Central Bank and market participants, and that there are no differences between the forecasts of market participants and the Central Bank given the same information set, all announced policy decisions that differ from the expectation of market participants very shortly before the announcements are plausible policy shocks, and all changes in financial markets prices around the same time window are a pure consequence of them. In the canonical papers by Gertler and Karadi (2015) and Nakamura and Steinsson (2018), these changes are referred to as "policy surprises", which are supposed to isolate purely exogenous policy perturbations (while the response of the Central Bank to the state of the economy is already captured by prices and by the expectations of market participants right before the announcements)[2].

Despite the increasing popularity of HFI methods, the necessary assumptions for surprises to serve as policy shocks are rarely met in practice. At least since Romer and Romer (2000), it has been widely accepted that the information set of the Central Bank with regards to the economy is different than that of market participants. Therefore, policy surprises include pure policy shocks combined with an "information revealing" effect. At

[1] See the chapter review by Ramey (2016).

[2] In the case of Colombia, Romero et al. (2021) are an example of the use of this technique to study monetary policy transmission. See also Kohlscheen (2014) for an example of the use of high-frequency changes in exchange rates around policy announcements to study the issue of the "exchange rate puzzle".

its core, this is what Ramey (2016) refers to as the "foresight problem": when the Central Bank reacts to its own expectations with regards to the state of the economy, and market participants have a different information set than that of the Central Bank, policy surprises are the sum of the pure policy shock and "news" about the expectation of the Central Bank with regards to key macroeconomic aggregates[3].

An example might help to reveal this problem. In January 2026, the Board of Directors of the Banco de la República (the Central Bank of Colombia) decided to increase its policy rate by 100 basis points (bps) as a response to inflationary concerns arising from a large Government-decreed increase in the nationwide minimum wage. Before the meeting of the Board, market analysts were, by and large, expecting a milder increase of 50 bps. After the policy announcement, if the policy surprise were a pure policy shock, inflation expectations should have fallen as an immediate response of the economy to a suddenly tighter stance of monetary policy. Instead, short-term inflation expectations (measured as the break-even between nominal and CPI-indexed government debt securities) increased significantly. This illustrates that the policy change contains not only a pure policy shock: due to its aggresiveness as a contractionary surprise, the announcement revealed to participants in financial markets news about the Central Bank's expectation for short-term inflation to which the policy decision in itself is reacting, which they duly incorporated in their forecasts.

Therefore, to properly estimate the transmission of monetary policy to the macroeconomy, it is essential to disentangle conventional monetary policy innovations from the central bank "information" shocks embedded within policy announcements. To achieve this, this paper builds on Jarociński and Karadi (2020) by exploiting high-frequency identification (HFI) through the comovement between short-term break-even inflation expectations (BEI) and overnight index swaps (OIS) around policy meetings of the Banco de la República. Standard New Keynesian theory (Galí, 2015) implies that an unexpected, contractionary policy shock lowers inflation expectations, whereas an information shock—signaling central bank optimism about economic fundamentals or impending inflationary pressures—increases them. Consequently, pure monetary policy shocks are identified by a negative comovement between high-frequency OIS and BEI adjustments, while central bank information shocks exhibit a positive comovement. These are integrated into an Instrumental Variable Bayesian Vector Autoregression (IV-BVAR) framework, following Cesa-Bianchi et

[3]Cieslak and Schrimpf (2019) estimate that these news are larger than the purely shock component in about 40% of policy announcements.

al. (2020), Gertler and Karadi (2015), Mertens and Ravn (2013), Miranda-Agrippino (2016), and Stock and Watson (2012). In this setup, the decomposed surprise series serve as external instruments to identify the structural columns of the VAR impact matrix. This econometric approach mitigates potential measurement error in high-frequency surprises, controls for the joint dynamic interactions among macroeconomic aggregates, and enables consistent estimation of dynamic impulse responses even when high-frequency instruments are available over a shorter time span than the underlying macroeconomic series.

The main findings of the paper can be summarized as follows. As has been observed in the literature, the response of the price level to a monetary policy surprise exhibits a "price puzzle" whereby a positive surprise induces an increase in the price level in the short term. This indicates that the response of the price level embeds the conventional contractionary effects of monetary policy with an additional informational component that induces an initial increase in prices. When estimating the response of the price level to pure policy shocks, the price puzzle disappears, according to standard theoretical predictions. The response of output is consistently negative under both scenarios, although volatile and less precisely estimated. Consistent with these results, the price level increases strongly in response to information shocks, an intuition already explained above. In this case, output exhibits a weaker response to the information shock.

By revisiting the effects of monetary policy on the economy, this paper contributes to three strands of the literature. From the more general to the more specific, the first strand focuses on the effects of monetary policy using different methods to isolate policy shocks. Key references include Christiano et al. (1999) and Romer and Romer (2004) using VAR-based methods, Gertler and Karadi (2015) using HFI methods, Ahmadi and Uhlig (2015) using sign restrictions, and Bernanke et al. (2005) using factor-augmented VAR methods. The second strand specializes on the use of HFI methods to isolate policy shocks. Besides the canonical works by Gertler and Karadi (2015), Nakamura and Steinsson (2018) and Cesa-Bianchi et al. (2020) (for the case of the UK), recent work by Bolhuis et al. (2024), Kohlscheen (2014) and Checo et al. (2024) deploys this method to estimate and use policy shocks for a large number of advanced and emerging economies. Finally, the third strand specifically focuses on observing that HFI policy surprises embed more than true policy shocks and on controlling for the information revealing effect of some policy surprises. On the observation that policy surprises include more than the policy shock itself, this paper is related to work by Miranda-Agrippino (2016), Nakamura and

Steinsson (2018), Bauer and Swanson (2022) and Cieslak and Schrimpf (2019). In controlling for information shocks, this paper is closely related to Jarociński and Karadi (2020), Miranda-Agrippino and Ricco (2021) and Pirozhkova et al. (2024). Our main contribution to this work is the direct use of market inflation expectations as a tool to isolate pure policy shocks from information shocks. Similar to Jarociński and Karadi (2020), this paper uses high-frequency financial market data, but in using directly inflation expectations, this paper avoids the use of other key prices (such as stock indices) which do not necessarily correlate with policy shocks as the theory would suggest (for instance, stock indices in Colombia are tightly related to global oil prices).

This paper unfolds as follows. Section 1 presents the two steps of the empirical strategy of this paper: the identification of policy and information shocks and the estimation of the effect of monetary policy on the macroeconomy. Section 2 briefly presents the data employed in the paper, while Section 3 presents the main results of the estimations. Some final thoughts are presented at the end as concluding comments.

1 Methodology

As explained before, this paper follows a two-step methodology. The first step consists in isolating policy shocks and information shocks using HFI on the comovement between break-even, short-term, market inflation expectations (BEI) and overnight index swaps (OIS) on the short-term interest rate around policy announcements by the Banco de la República. This step follows closely Cieslak and Schrimpf (2019) and Jarociński and Karadi (2020). The second step is the estimation of the dynamic response of output and inflation to these shocks using a standard IV-BVAR approach.

1.1 Policy Surprises, Policy Shocks and Information Shocks

This paper measures monetary policy surprises using Overnight Index Swaps (OIS) on the IBR (*Indicador Bancario de Referencia*), the key market reference for a credit risk-free overnight rate in Colombia. These contracts involve the exchange of a fixed interest rate for a floating rate determined by the compounded overnight IBR over a given maturity. As a result, the fixed rate embedded in the contract summarizes market expectations about the future path of short-term interest rates at the time of pricing. This paper uses the the one-month IBR swap rate set before the monetary policy meeting as the market expectation of the short-term future policy rate i_t^M at time t, $E[i_t^M \mid t-\delta]$. Here, $t-\delta$ denotes the information set available to market participants shortly before the policy announcement.

In developed economies, HFI identification of pure policy shocks typically relies on narrow windows around monetary policy announcements (Cesa-Bianchi et al., 2020; Gertler & Karadi, 2015). In these settings, δ is often defined as a 30-minute window around the announcement (Nakamura and Steinsson (2018)). However, this approach is generally not feasible in emerging economies, where financial markets are often closed when the policy decision is announced (Kohlscheen, 2014). In the case of the Colombian OIS market considered here, the relevant information becomes available on the day the OIS-implied rate for the following month is revealed, which occurs before the monetary policy meeting.

Monetary policy surprises are then defined as the difference between the realized policy rate and the market-implied expected rate:

$$MPS_t = i_t^M - E[i_t^M \mid t-\delta].$$

In this specification, MPS_t captures the unanticipated component of the monetary policy announcement.

As explained above, MPS_t is likely to embed, beyond exogenous policy innovations, news or information shocks with regard to the future state of the economy. The latter might reflect, for instance, an informational or technological advantage held by the Central Bank in the construction of relevant forecasts, and is partially revealed when the policy decision is announced (Bauer & Swanson, 2023; Miranda-Agrippino, 2016; Nakamura & Steinsson, 2018). To isolate these two components, this paper decomposes MPS_t into a pure monetary policy shock and a Central Bank information shock, using an identification strategy closely related to Cieslak and Schrimpf (2019) and Jarociński and Karadi (2020). While those papers exploit the joint high-frequency response of interest rates and stock prices around policy announcements, this paper instead uses the joint response of monetary policy surprises and short-term, market, break-even, inflation expectations. The underlying logic is that high-frequency financial-market reactions reveal whether the announcement is interpreted primarily as a policy innovation or as news about fundamentals (Jarociński & Karadi, 2020).

More specifically, if MPS_t captured only a pure monetary policy shock, an unexpected tightening should be followed by a decline in market inflation expectations. Such a response would reflect the standard transmission mechanism of a standard, New Keynesian, model: persistently tighter policy reduces expected demand and expected inflationary pressures. Conversely, if inflation expectations increase after the announcement, the surprise cannot be understood as purely monetary. Rather, it indicates that the central bank has revealed information consistent with stronger inflationary pressures than those previously priced by the market. In that case, the observed surprise reflects plausibly a contractionary policy coherent with an information effect working in the opposite direction on expectations. This observation provides the basis for this decomposition: the high-frequency comovement of MPS_t and market inflation expectations allows the isolation of pure monetary policy shocks and central bank information shocks.

Market inflation expectations are measured using breakeven inflation (BEI), defined as the difference between nominal TES bond yields and real UVR-indexed bond yields at a given horizon, h. BEI is a key market-based indicator for assessing the degree of anchoring of long-term inflation expectations. The change in BEI around the announcement, ΔBEI_t, provides the high-frequency change in inflation expectations used in the methodology described below.

$$\Delta BEI_t = E[\pi_{t+h} \mid t+\delta] - E[\pi_{t+h} \mid t-\delta].$$

In the empirical implementation, $t+\delta$ corresponds to the business day immediately after the announcement, while $t-\delta$ corresponds to the business day immediately before the announcement, and h is set at 1 year.[4] Figure 1 presents a scatterplot of MPS_t and ΔBEI_t. The figure shows that, for some policy announcements, unexpected positive monetary surprises are accompanied by increases in market-based inflation expectations. These observations are interpreted as evidence that the monetary policy surprise contains a relevant central bank information component.

Using these two high-frequency measures, this paper implements the procedure proposed by Jarociński and Karadi (2020). This procedure generates two orthogonal shocks: one associated with the central bank information component, u_t^i, and another associated with pure monetary policy shocks,

[4]BEI may in principle contain information other than inflation expectations, such as inflation risk premia and liquidity premia (Espinosa-Torres et al., 2017) which are not cleaned up in this paper.

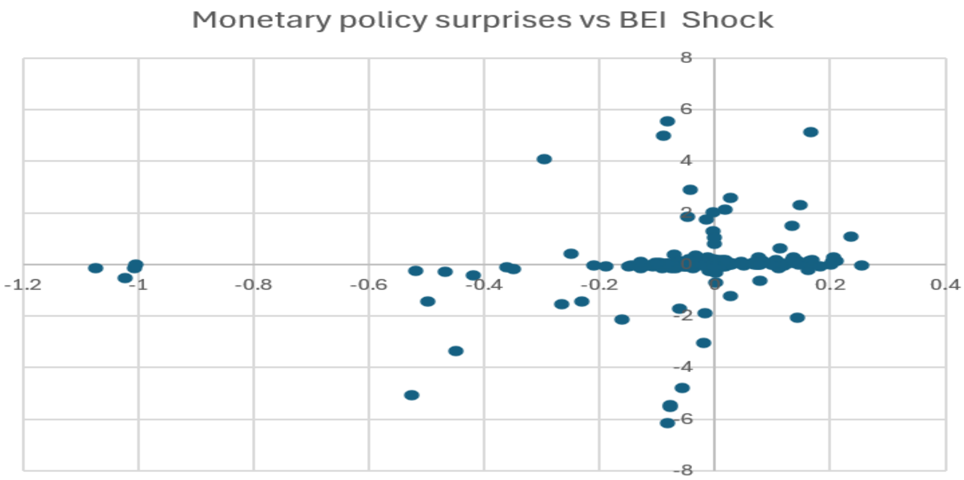

Figure 1: Scatter plot of high frequency movements in BEI and Monetary Policy Surprises. *Note*: Scatter plot of MPS_t and ΔBEI_t around monetary policy announcements.

u_t^m. Identification relies on the information embedded in ΔBEI_t and follows the sign restrictions of Table 1. Specifically, u_t^i is expected to be active whenever there is a positive comovement between ΔBEI_t and MPS_t, whereas u_t^m is expected to be active when the comovement is negative. A complete description of the methodology is presented in Appendix A.

Variable	u_t^m	u_t^i
MPS_t	$+$	$+$
ΔBEI_t	-	$+$

Table 1: Sign restrictions for identification.

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Note: Sign restrictions to identify central bank information componetn (u_t^i) and raw monetary policy component (u_t^m).

1.2 The Effects of Monetary Policy

To estimate the dynamic effects of unexpected monetary policy innovations and information shocks on macroeconomic variables, this paper employs an Instrumental Variable Bayesian Vector Autoregression (IV-BVAR) framework, following Gertler and Karadi (2015), Mertens and Ravn (2013), and Stock and Watson (2012). Rather than treating the high-frequency surprises as direct measures of structural shocks, they are used as external instruments to identify the structural responses from the reduced-form residuals.

Consider the standard VAR(p) representation:

$$Y_t = \sum_{j=1}^{p} B_j Y_{t-j} + u_t, \tag{1}$$

where Y_t is an $(n \times 1)$ vector of macroeconomic variables, B_j are $(n \times n)$ coefficient matrices, and u_t is the vector of reduced-form innovations with covariance matrix $\Sigma_u = \mathbb{E}[u_t u_t']$. The reduced-form residuals are assumed to be a linear combination of mutually orthogonal structural shocks ε_t:

$$u_t = A\varepsilon_t, \quad \text{with} \quad \Sigma_u = AA', \tag{2}$$

where A is the structural impact matrix. The model is estimated using Bayesian techniques with a standard Normal-Inverted Wishart (NIW) prior, which mitigates the curse of dimensionality while preserving dynamic relationships.

To uniquely identify the dynamic response of the system to a specific structural shock $\varepsilon_{1,t}$ (e.g., the monetary policy shock), let Z_t denote the corresponding external instrument. The instrument must satisfy the standard relevance and exogeneity conditions (Stock & Watson, 2018):

$$\mathbb{E}[\varepsilon_{1,t} Z_t] = \alpha \neq 0 \quad \text{and} \quad \mathbb{E}[\varepsilon_{2:n,t} Z_t] = 0. \tag{3}$$

Under the unit-effect normalization ($A_{1,1} = 1$), the covariance between the reduced-form innovations and the instrument identifies the relative impact responses (the first column of A, denoted $A_{1:n,1}$):

$$\frac{\mathbb{E}[u_{i,t} Z_t]}{\mathbb{E}[u_{1,t} Z_t]} = A_{i,1}. \tag{4}$$

Empirically, the impact vector $A_{1:n,1}$ is obtained by performing an instrumental variable regression of the reduced-form residuals of all variables $u_{i,t}$ on the reduced-form residual of the policy interest rate $u_{1,t}$, using Z_t as the instrument. Using the posterior draws of the autoregressive lag polynomial $\hat{B}(L)$, the structural Impulse Response Functions (IRFs) are computed as:

$$\text{IRF}_h = \left(I_n - \sum_{j=1}^{p} \hat{B}_j L^j \right)^{-1} A_{1:n,1}. \tag{5}$$

This identification strategy is carried out separately for the raw interest rate surprise MPS_t, the pure monetary policy shock u_t^m, and the information shock u_t^i. A major practical advantage of the IV-BVAR approach is that the instrument Z_t does not need to cover the entire historical estimation sample of Y_t; consistent identification of $A_{1:n,1}$ requires only overlapping periods between the VAR residuals and the high-frequency surprises.

The BVAR model is estimated using a standard conjugate Normal–Inverse-Wishart prior, implemented via dummy observations following the Minnesota tradition (Bańbura et al., 2010). Because the endogenous variables enter the system in levels, the prior mean on the first own lag is set to $\delta = 1$, which centers the prior on a random-walk process suitable for persistent macroeconomic aggregates. The overall degree of shrinkage is governed by the hyperparameter $\lambda = 0.2$, which controls the prior variance of the autoregressive parameters across lag orders $l = 1, \ldots, p$. This penalizes cross-variable lags and shrinks higher-order coefficients toward zero at a quadratic decay rate. To accommodate potential cointegration relationships and common stochastic trends among the non-stationary series, the system is augmented with a sum-of-coefficients prior with a proportional tightness of $\tau = 10\lambda = 2.0$. The prior on the deterministic constant is uninformative, specified with a diffuse tightness parameter of $\epsilon = 10^{-3}$ so that the unconditional sample means are determined directly by the data. Finally, the scale parameters, σ_i^2, which govern the prior covariance matrix, are calibrated internally using the residual variances from univariate AR(p) regressions estimated for each series.

2 Data

The benchmark system for the estimation of the parameters in the BVAR includes four observables: the consumer price index (CPI), the core CPI, the logarithm of GDP, and the interbank interest rate (TIB). The CPI corresponds to the monthly consumer price index published by the National Administrative Department of Statistics in Colombia (*DANE*, for its Spanish acronym). The core CPI corresponds to the non-energy non-food consumer price index obtained from *Banco de la República*. Since GDP is only available at quarterly frequency, we construct a monthly GDP series by temporally disaggregating the official quarterly real GDP using the monthly economic activity indicator (*ISE*) produced by *DANE* as an auxiliary indicator. To do so, we use the procedure in Sax and Steiner (2013), following the methodology of Chow and Lin (1971). In addition, we include the monthly average of interbank interest rate, which closely tracks the behavior of the monetary policy rate. Data are monthly and span the period from 2008:1 to 2025:12.

High-frequency inflation expectations and policy surprises are constructed as explained above and relevant information is sourced from the Banco de la República's Market Analysis Department. Data span from 2009:1 to 2025:12.[5]

3 Results

3.1 Baseline Results

This subsection presents the results of estimating the system 1 and identifying responses to monetary policy using the three alternative instruments: namely MPS_t, u_t^m, and u_t^i. Figure 2 reports the responses of the macroeconomic variables after a 25 basic points increase in the interest rate, when the shock is instrumented by the monetary policy surprise, MPS_t.[6] The response of headline CPI displays a non-monotonic pattern. In the short run, prices increase slightly and reach a peak of approximately 0.10% around month 6, reflecting the mild price puzzle frequently documented in the high-frequency monetary policy literature. Subsequently, the response turns persistently negative after horizon 15, reaching a trough of about -0.35% by month 36. In contrast, core inflation (CPISAR) declines almost immediately, exhibiting a small initial contraction of -0.10% before steadily falling to roughly -0.70% toward the end of the horizon.

Output (GDP) contracts on impact by approximately -0.50%, followed by a mild short-lived rebound before deepening into a persistent trough of around -0.55% between months 12 and 18, and slowly reverting thereafter. Overall, Figure 2 indicates that while the surprise MPS_t induces standard contractionary dynamics across real activity and core prices, it also captures transient headline price increases.

Figure 3 reports the same exercise using u_t^m, the pure monetary policy component obtained from the decomposition, as an external instrument. Several key features stand out. First, the short-run increase in headline CPI observed with the raw surprise completely disappears. Prices decline monotonically throughout the horizon, reaching a trough of approximately -0.55% around month 35. Similarly, core inflation (CPISAR) contracts

[5]We restrict the sample to 2009 onward because TES UVR yields during the 2008 global financial crisis exhibited heightened volatility driven primarily by liquidity constraints and market distress rather than shifts in inflation expectations.

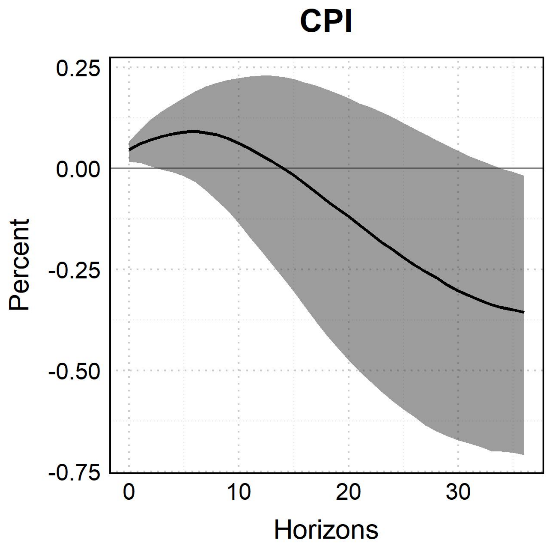
[6]The first-stage F-statistic (9.54 in this estimation) tests the instrument's relevance by measuring its correlation with the target reduced-form residual. This value is just below the conventional strong-instrument threshold of 10, indicating the instrument is informative but bordering on weak.

immediately from an impact response of -0.12% to a persistent trough of around -0.90% after horizon 30. This pattern aligns directly with standard theoretical predictions, demonstrating that removing the information component fully mitigates the price puzzle.

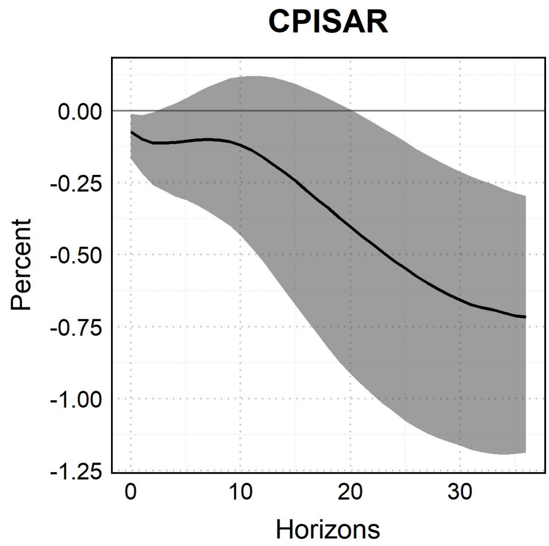
Second, output (GDP) contracts sharply on impact by about -1.15%, followed by a quick rebound within the first five months toward -0.85%, and subsequently stabilizes with a gradual recovery toward -0.65% by month 36. Taken together, these findings confirm that isolating the pure monetary policy innovation yields contractionary transmission dynamics that are fully consistent with standard macroeconomic theory.

Finally, Figure 4 presents the impulse responses to an unexpected interest rate increase instrumented with u_t^i, the central bank information component. The response of headline CPI rises on impact to 0.06% and reaches a peak of approximately 0.14% around month 7. This confirms that the transient short-run increase in prices observed under the raw surprise (MPS_t) is primarily driven by this information channel. After horizon 18, the headline price response turns negative, reaching a trough of about -0.28% by month 36. Core inflation (CPISAR) remains relatively subdued in the short run—hovering between -0.08% and -0.03% during the first 10 months—before steadily declining to roughly -0.63% at longer horizons.

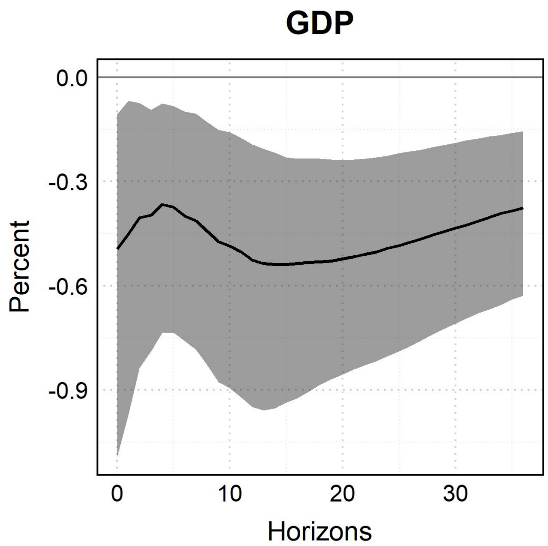
Output (GDP) displays a milder contraction on impact compared to the pure monetary shock, falling by around -0.21% and remaining stable for the first 5 months. It then contracts further toward a trough of -0.39% around horizon 17 before gradually recovering to -0.26% by month 36. Overall, the comparative evidence across Figures 2–4 demonstrates that the raw monetary policy surprise (MPS_t) conflates two distinct forces: a conventional monetary tightening shock and an endogenous information shock. Once decomposed and identified through the IV-BVAR framework, the pure monetary policy shock (u_t^m) generates unambiguous contractions in output and prices without inducing a price puzzle, while the information component (u_t^i) accounts for the short-run inflationary pressures.



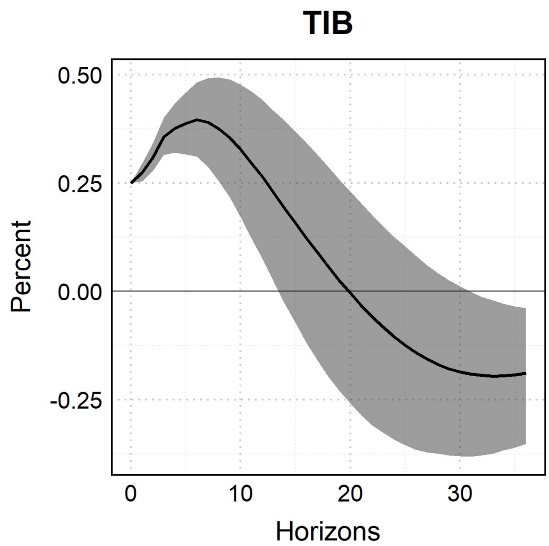
(a) IRF of CPI.



(b) IRF of core CPI.



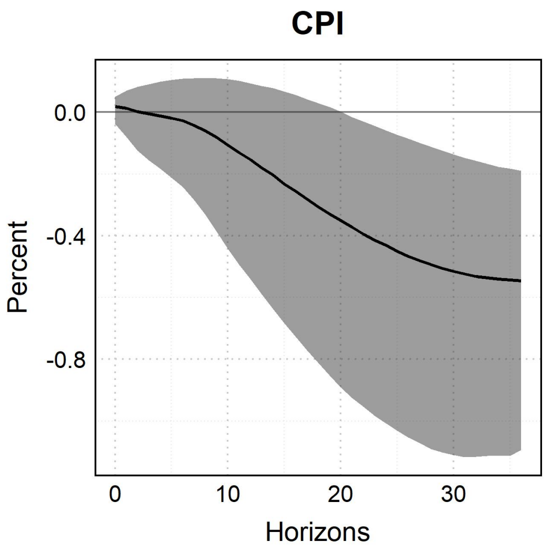
(c) IRF of GDP.



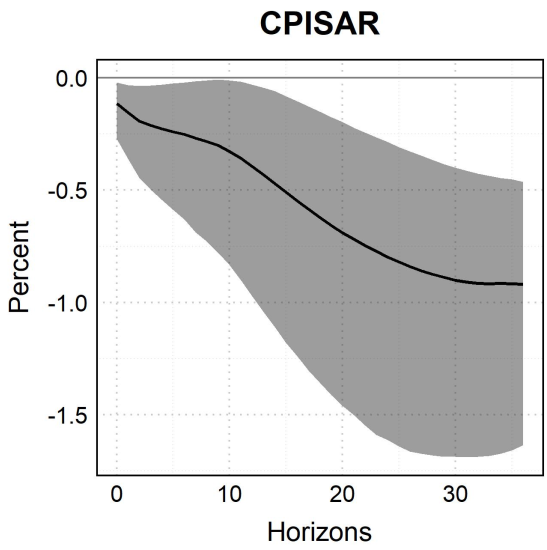
(d) IRF of GDP.

Figure 2: IRFs to an unexpected increase in the interest rate using the MPS_t as instrument.

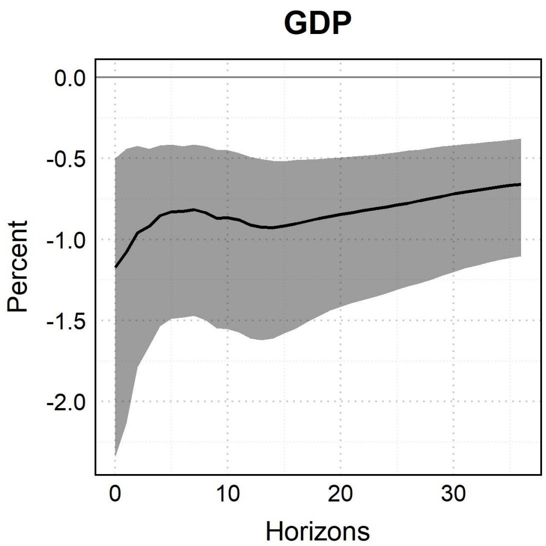
Note: Macroeconomic responses to monetary policy shock instrumented with MPS_t. Panel (a) reports the impulse response of CPI, panel (b) reports the impulse response of core CPI, panel (c) reports the impulse response of GDP, and panel (d) reports the impulse response of TIB. Shaded areas are 68.0% posterior bands.



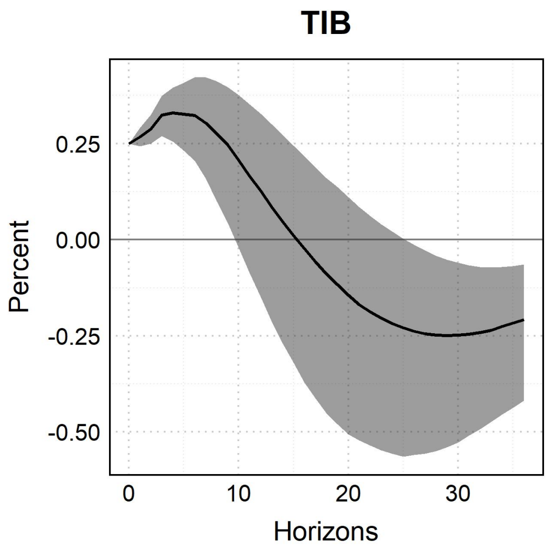
(a) IRF of CPI.



(b) IRF of core CPI.



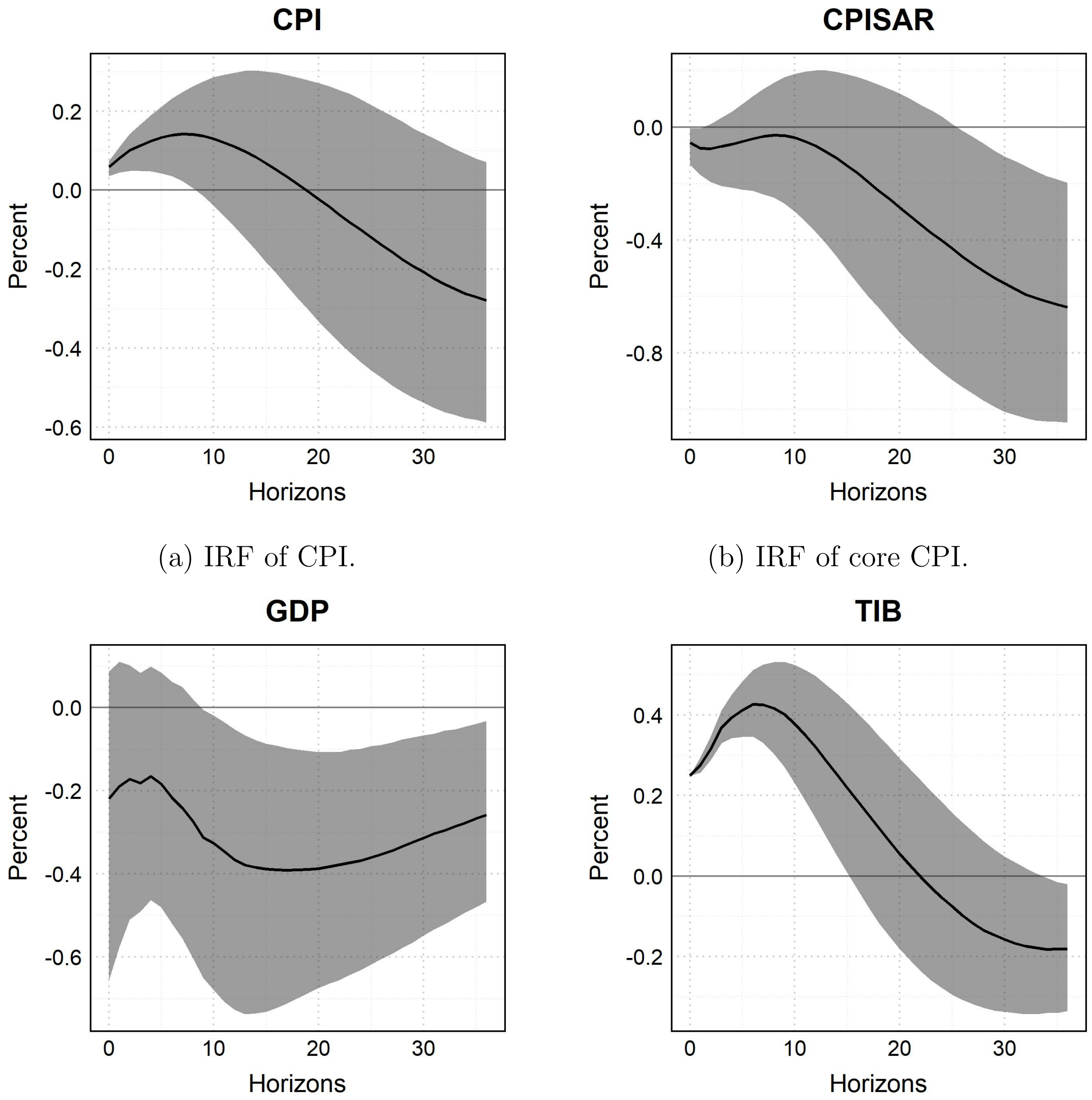
(c) IRF of GDP.



(d) IRF of GDP.

Figure 3: IRFs to an unexpected increase in the interest rate using the u_t^m as instrument.

Note: Macroeconomic responses to monetary policy shock instrumented with u_t^m, the raw monetary policy component. Panel (a) reports the impulse response of CPI, panel (b) reports the impulse response of core CPI, panel (c) reports the impulse response of GDP, and panel (d) reports the impulse response of TIB. Shaded areas are 68.0% posterior bands.



(a) IRF of CPI. (b) IRF of core CPI.

(c) IRF of GDP. (d) IRF of GDP.

Figure 4: IRFs to an unexpected increase in the interest rate using the u_t^i as instrument.

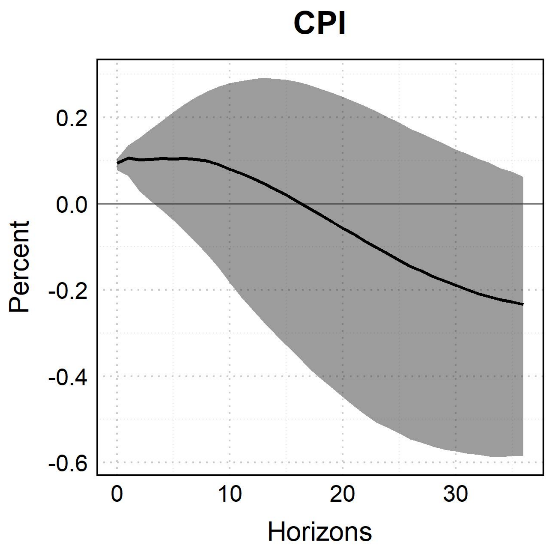
Note: Macroeconomic responses to monetary policy shock instrumented with u_t^i, the central bank information component. Panel (a) reports the impulse response of CPI, panel (b) reports the impulse response of core CPI, panel (c) reports the impulse response of GDP, and panel (d) reports the impulse response of TIB. Shaded areas are 68.0% posterior bands.

3.2 Identification with Stock Prices

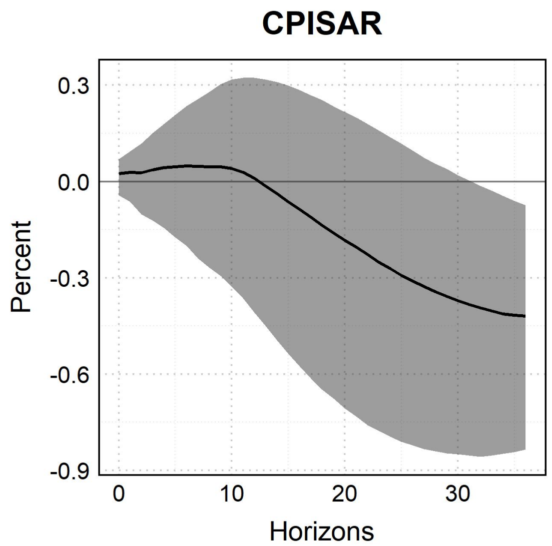
An alternative approach to identifying the central bank information component relies on the high-frequency response of stock market prices around monetary policy announcements. This strategy follows Jarociński and Karadi (2020), who argue that equity prices provide a useful signal to distinguish between conventional monetary policy shocks and information effects. The underlying intuition is straightforward. Under standard asset-pricing logic, an unexpected monetary tightening should lower stock market valuations, since higher interest rates raise discount rates and are typically associated with weaker expected future activity. By contrast, if stock prices increase following a positive monetary policy surprise, this suggests that the announcement may have also revealed favorable information about the economic outlook. In this subsection, we implement this alternative identification strategy for Colombia using the reaction of the COLCAP index, the benchmark stock market index for the Colombian economy. The response of the COLCAP around monetary policy announcements provides an alternative high-frequency signal to isolate the information content embedded in monetary policy surprises.

The decomposition based on the high-frequency response of the COLCAP leads to results that appear less consistent with the evidence obtained from break-even inflation. In the BEI-based exercise, the identifying variable is directly related to inflation expectations, so the information component captures the part of the monetary policy surprise associated with revisions in expected inflation. By contrast, the COLCAP-based decomposition relies on a different financial-market signal, reflecting channels that are not directly related to the inflation-expectations mechanism emphasized in this paper.

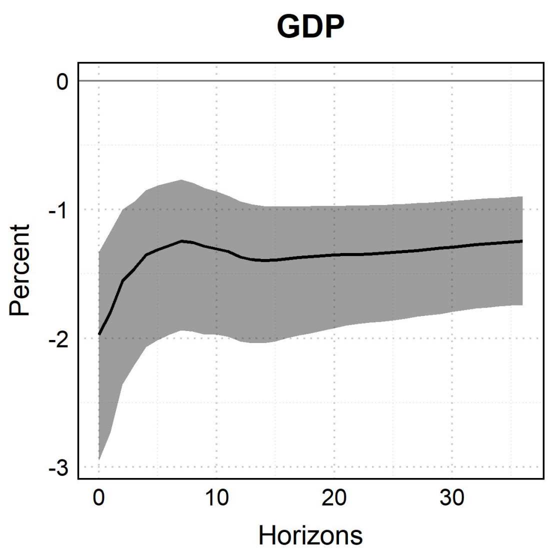
This distinction is reflected in the estimated responses. The COLCAP-based decomposition reallocates the price response across shocks in a way that differs from the BEI-based exercise and from the pattern emphasized by Jarociński and Karadi (2020). In the BEI-based decomposition, the central bank information component accounts for the initial increase in CPI, consistent with the interpretation that monetary policy announcements convey information about inflationary pressures. By contrast, when COLCAP is used as the financial-market signal, the price puzzle instead appears to be concentrated in the pure monetary policy component: CPI increases after the monetary policy shock during the first horizons before declining later (Figure 5).The CPI response to the information component is instead relatively muted at short horizons and becomes negative over the medium run (Figure 6). Hence, using COLCAP does not reproduce the same separation found in Jarociński and Karadi (2020), where a positive co-movement between interest rate surprises and stock prices is interpreted as favorable central bank information and is associated with higher real activity and prices. Rather, in the Colombian application, the stock-market-based decomposition appears to assign the puzzling short-run price increase mainly to the monetary policy component, suggesting that stock market valuations may provide a noisier signal than break-even inflation for identifying the inflation-expectations channel.



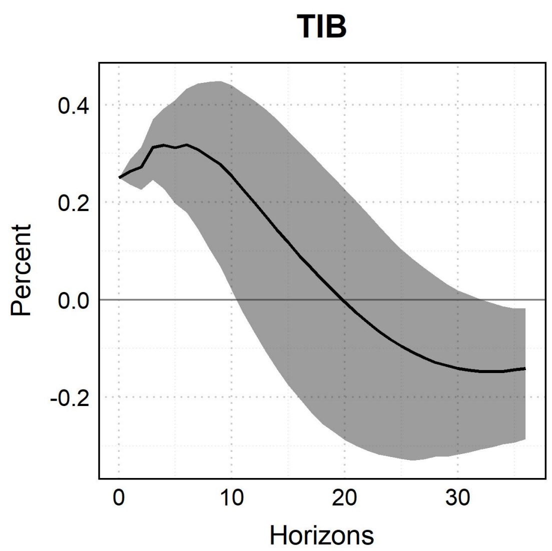
(a) IRF of CPI.



(b) IRF of core CPI.



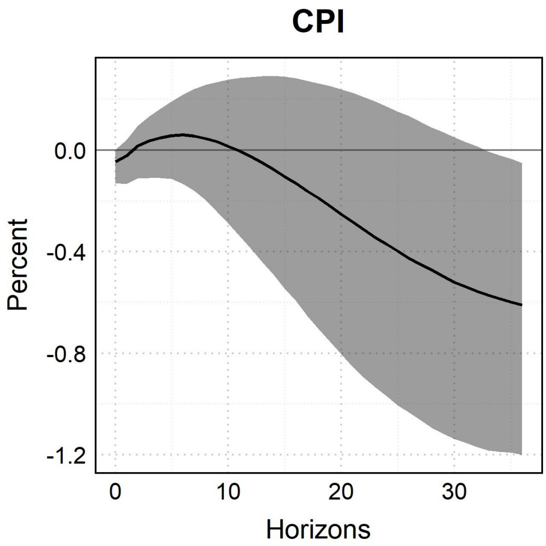
(c) IRF of GDP.



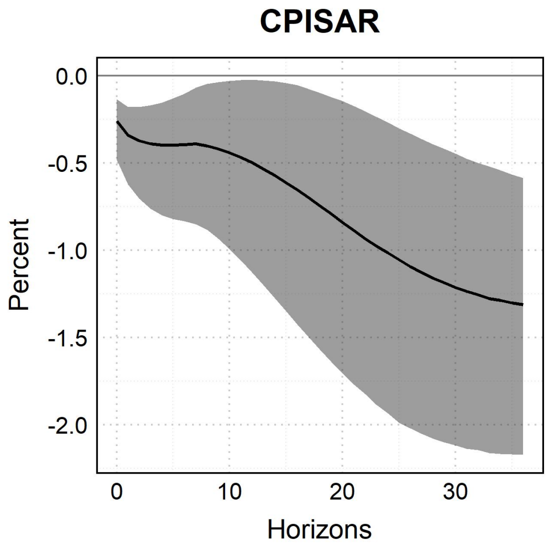
(d) IRF of GDP.

Figure 5: IRFs to an unexpected increase in the interest rate using the u_t^m as instrument.

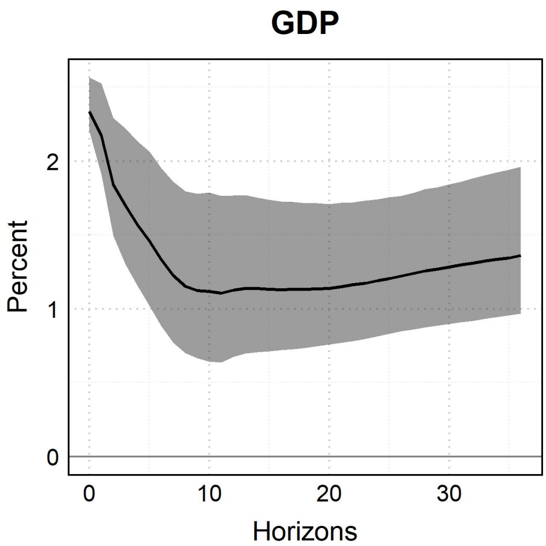
Note: Macroeconomic responses to monetary policy shock instrumented with u_t^m, the raw monetary policy component. When decomposing the surprises, we use COLCAP changes to decompose the surprises. Panel (a) reports the impulse response of CPI, panel (b) reports the impulse response of core CPI, panel (c) reports the impulse response of GDP, and panel (d) reports the impulse response of TIB. Shaded areas are 68.0% posterior bands.



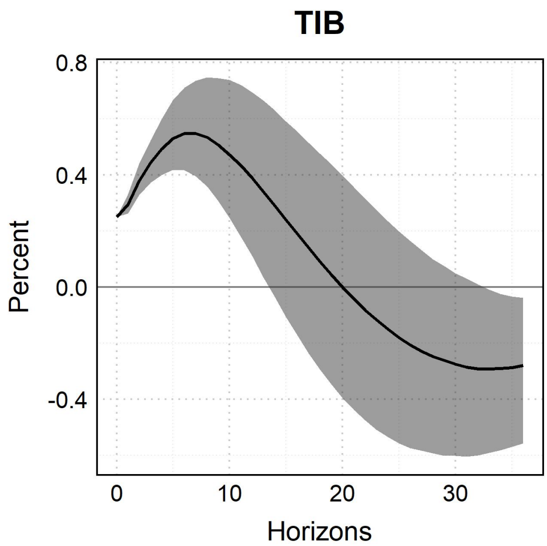
(a) IRF of CPI.



(b) IRF of core CPI.



(c) IRF of GDP.



(d) IRF of GDP.

Figure 6: IRFs to an unexpected increase in the interest rate using the u_t^i as instrument.

Note: Macroeconomic responses to monetary policy shock instrumented with u_t^i, the central bank information component. When decomposing the surprises, we use COLCAP changes to decompose the surprises. Panel (a) reports the impulse response of CPI, panel (b) reports the impulse response of core CPI, panel (c) reports the impulse response of GDP, and panel (d) reports the impulse response of TIB. Shaded areas are 68.0% posterior bands.

These results suggest that the COLCAP may not be the most appropriate financial variable to identify the central bank information component of monetary policy surprises in the Colombian case. One possible explanation is that movements in the COLCAP are strongly influenced by fluctuations in international oil prices. This is particularly relevant for Colombia because the stock market index has historically included a sizable participation of firms whose valuations are directly or indirectly exposed to the oil sector. As a result, changes in the COLCAP around monetary policy announcements may reflect not only the information revealed by the central bank, but also contemporaneous movements in global commodity prices that are unrelated to domestic monetary policy.

This concern is illustrated in Figure 7, which reports the annual percentage growth of the COLCAP and the West Texas Intermediate (WTI) oil price over the sample period. The two series display a strong positive association, with a correlation of 0.60, suggesting that Colombian stock market valuations are highly sensitive to oil price dynamics. This dependence is also present at higher frequencies. When we compute the change in WTI around monetary policy meetings, which should be exogenous to the information released by the Colombian central bank, its correlation with the high-frequency change in the COLCAP is 0.29. This value is economically relevant and indicates that part of the stock market reaction observed around policy announcements may be capturing global oil price movements rather than domestic monetary policy news. By contrast, the correlation between the high-frequency change in WTI and the high-frequency change in break-even inflation is only -0.08, suggesting that BEI is substantially less exposed to this external commodity-price factor.

This evidence provides a plausible explanation for why the COLCAP-based decomposition delivers a less clear separation between the monetary policy component and the central bank information component. If the stock market response partly reflects oil price movements, then the identifying signal used in the decomposition is contaminated by an external shock that is orthogonal to the monetary policy announcement itself. In that case, a positive co-movement between monetary policy surprises and the COLCAP cannot be interpreted exclusively as favorable central bank information about the domestic economic outlook. It may instead reflect the effect of oil price changes on the valuation of oil-related firms, on expected export revenues, or on broader perceptions about the Colombian economy as a commodity-exporting country. Therefore, the COLCAP-based signal may assign part of the observed price dynamics to the wrong structural component, generating misleading impulse responses.

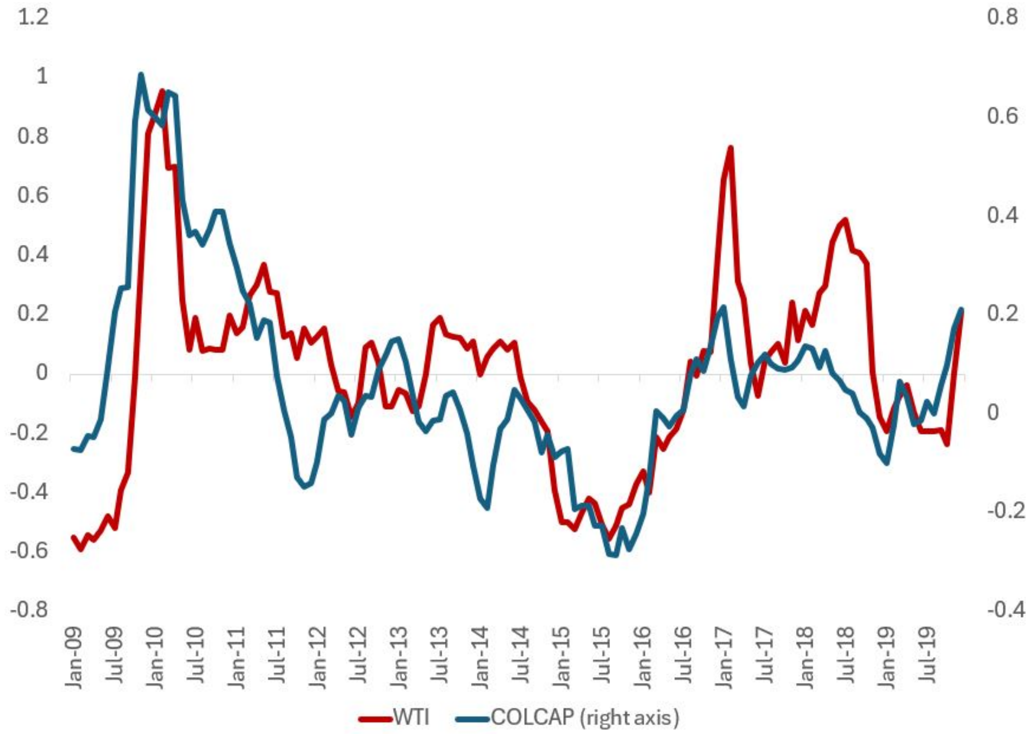


Figure 7: Annual growth of WTI and COLCAP

Note: The figure reports the annual percentage growth of the West Texas Intermediate oil price and the COLCAP index over the sample period. The COLCAP series is plotted on the right axis. The positive co-movement between the two series suggests that Colombian stock market valuations are strongly exposed to international oil price dynamics.

This interpretation is consistent with the results reported above. Unlike the BEI-based decomposition, which directly exploits revisions in inflation expectations, the COLCAP-based decomposition relies on a financial variable that is affected by multiple forces, including discount rates, expected profits, risk premia, external financial conditions, and commodity prices. The strong association between the COLCAP and WTI suggests that the stock market signal is not clean enough to isolate the inflation-expectations channel emphasized in this paper. Hence, the fact that the price puzzle appears mainly in the monetary policy component when using COLCAP should not be interpreted as evidence against the central bank information channel. Rather, it suggests that, in the Colombian context, the stock market response around policy announcements may be too strongly influenced by oil price movements to provide a reliable measure of central bank information.

Concluding Comments

Consistent with evidence presented elsewhere (Jarociński & Karadi, 2020; Miranda-Agrippino, 2016), this paper has shown that the separation between information shocks and policy shocks contained in traditional HFI monetary policy surprises allows us to estimate the effects of monetary policy more precisely obtaining responses that are closer to theoretical predictions and to basic economic intuitions.

This paper reflects the outcomes of research work in progress, and several avenues for future work will be considered to sttrengthen the analysis. Critically, the change in break-even inflation expectations calculated here uses a time window of two business days (policy announcements plus and minus one day). This can be refined by employing a shorter time window of half a day, capturing inflation expectations very shortly before policy announcements and right at the beginning of the day after the policy announcements. This implies a different set of data requirements which should be overcome in the near future.

In addition to this, the estimation of pure policy shocks that are robust to informational and technological advantages on the side of the Central Bank will allow us to revisit key questions in the literature on the effects of monetary policy, with regards, for instance, to the transmission of monetary policy to credit market interest rates and to the effect of the institutional context on that transmission.

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Appendices

A Orthogonalization of surprises

Following Jarociński and Karadi (2020), let

$$M_t = \begin{bmatrix} MPS_t & \Delta BEI_t \end{bmatrix}$$

denote the vector of high-frequency surprises, where MPS_t is the monetary policy surprise and ΔBEI_t is the change in break-even inflation around the announcement. We assume that, within the announcement window, these surprises are driven only by two structural shocks, so that

$$M_t = U_t C, \qquad U_t = \begin{bmatrix} u_t^m & u_t^i \end{bmatrix},$$

where u_t^m denotes the conventional monetary policy shock and u_t^i denotes the central bank information shock. The matrix C collects the contemporaneous effects of these shocks on the observed high-frequency surprises.

To construct the orthogonal components, the procedure first apply a QR decomposition to the matrix of observed surprises,

$$M = QR,$$

where $Q'Q = I$. The QR decomposition provides an orthogonal basis for the two-dimensional space spanned by the observed surprises. However, these orthogonal components do not yet have an economic interpretation. Therefore, the method rotate this basis using an orthogonal rotation matrix

$$P(a) = \begin{bmatrix} \cos(a) & \sin(a) \\ -\sin(a) & \cos(a) \end{bmatrix},$$

where the angle a is chosen so that the resulting impact matrix satisfies the required sign restrictions. In particular, after the rotation we obtain

$$U = QP(a)D, \qquad C = D^{-1}P(a)'R,$$

where D is a diagonal scaling matrix. This normalization implies that the monetary policy surprise can be additively decomposed as

$$MPS_t = u_t^M + u_t^I.$$

with the identifying sign restrictions

$$c_M < 0, \qquad c_I > 0.$$

Thus, a conventional monetary policy shock is identified as an unexpected tightening that raises the monetary policy surprise and reduces break-even inflation, while a central bank information shock is identified as an unexpected tightening associated with an increase in break-even inflation. The QR decomposition is therefore used only to extract orthogonal components from the observed surprise space; the economic interpretation is obtained after rotating those components so that the sign restrictions are satisfied.